

Introduction to lattice based cryptography

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September 2026, Barcelona

Motivation

- till now most cryptography relied on hardness of finding prime factors or finding the discrete logarithm, ie. RSA, Elliptic Curves, etc
- Shor's algorithm suggests that factorization can be done in polynomial time (on quantum computers).
 - given an odd composite $n \in \mathbb{Z}$, find $d \in \mathbb{Z}$, such that $1 < d < n$ and $d|n$.

Status of the ecosystem

- isogenesis based
- hash based
- codes based
- lattice based
- multivariate based

TODO

Motivation

- Ethereum, is going with hash-based signatures for the validators.
 - aggregation through SNARK proofs (eg. binius, flock, etc)
 - each slot a different XMSS leaf
 - the most conservative choice in terms of security
- hash-based simpler to implement, less potential errors.
Whereas lattice-based have more subtelties
- hash-based schemes don't have much algebraic structure
 - good for less attacks, not so fun to build stuff with
 - eg. elliptic curves have algebraic structure, allows for signature aggregation, thresholds, etc
- lattice-based schemes have smaller proofs, signatures and public keys

Rings

Definition (Ring)

A set A with operations called *addition* and *multiplication* which satisfy the following axioms:

- i. A with addition alone is an abelian group.
- ii. Multiplication is associative.
- iii. Multiplication is distributive over addition. That is, $\forall a, b, c \in A$,

$$a(b + c) = ab + ac$$

$$(b + c)a = ba + ca$$

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$I \subset R$ (R ring) such that $0 \in I$ and $\forall x \in I, r \in R, xr, rx \in I$.

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Eg: Ring $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, 3, \dots\}$, Ideal $n\mathbb{Z} = \{nk \mid k \in \mathbb{Z}\}$ for fixed $n \in \mathbb{Z}$.

Vector spaces

Definition (Vector space)

A *vector space* over a field F is a set V , with two operations $+$, $\cdot$, called *vector addition* and *scalar multiplication*, such that

- V with vector addition is an abelian group
- $\forall k \in F$ and $\vec{a} \in V$, the scalar product $k\vec{a}$ is an element of V , subject to the following conditions: $\forall k, l \in F, \vec{a}, \vec{b} \in V$
 - i. $k(\vec{a} + \vec{b}) = k\vec{a} + k\vec{b}$
 - ii. $(k + l)\vec{a} = k\vec{a} + l\vec{a}$
 - iii. $k(l\vec{a}) = (kl)\vec{a}$
 - iv. $1\vec{a} = \vec{a}$
- *basis*: set of vectors that *span* the vector space

Modules

Definition (Module)

Let A be a ring. An A -module is an Abelian group M with a multiplication map

$$\begin{aligned} A \times M &\longrightarrow M \\ (f, m) &\longmapsto fm \end{aligned}$$

satisfying $\forall f, g \in A, m, n \in M$.

- i. $f(m \pm n) = fm \pm fn$
- ii. $(f \pm g)m = fm \pm gm$
- iii. $(fg)m = f(gm)$
- iv. $1_A m = m$

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Remark: Vector spaces are defined over fields, a Module is the generalization of a vector space where the scalars come from a ring instead of a field.

Lattices

Definition (Lattice)

A lattice $\mathcal{L}$ in $\mathbb{R}^n$ is the set of all integer linear combinations of m linearly independent vectors $b_1, \dots, b_m \in \mathbb{R}^n$, with $m \leq n$.

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$B = b_1, \dots, b_m \in \mathbb{R}^n$ is the *basis* of $\mathcal{L}$. We denote it $\mathcal{L} = \mathcal{L}(B)$.

$$\mathcal{L}(B) := B \cdot \mathbb{R}^n = \left\{ \sum_{i=1}^n c_i b_i \mid c_i \in \mathbb{R}^n \right\}$$

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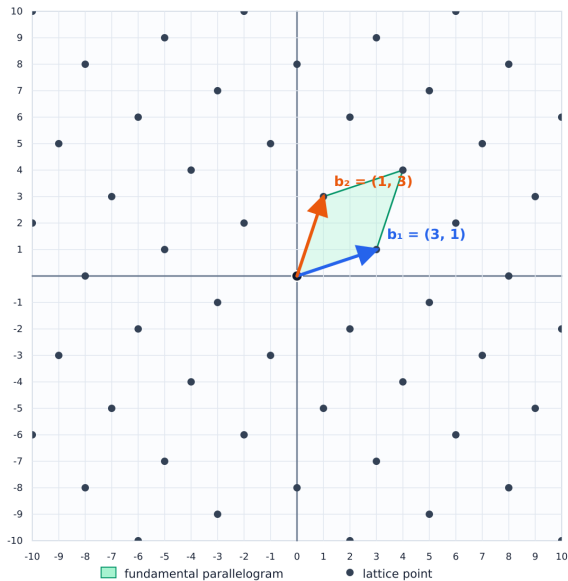
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In most cryptographic applications, $b_1, \dots, b_n \in \mathbb{Z}^n$, ie. $\mathcal{L}$ is an integer lattice.

2-dimensional lattice

$$\Lambda = \{n b_1 + m b_2 : n, m \in \mathbb{Z}\}, b_1 = (3, 1), b_2 = (1, 3)$$



Lattice properties

a n -dimensional lattice $\mathcal{L}$ is

- i. *additive subgroup* of $\mathbb{R}^n$: $\mathcal{L}$ is the nonempty subset of $\mathbb{R}^n$ closed under vector addition (and subtraction),

$$\text{ie. } \forall x, y \in \mathcal{L}, -x, x + y \in \mathcal{L}$$

- ii. *discrete*: every $x \in \mathcal{L}$ has a neighborhood in $\mathbb{R}^n$ in which x is the only lattice point.

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Successive minima $\lambda_i(\mathcal{L})$: smallest radius r s. th. $\mathcal{L}$ the n -dimensional ball of radius r contains i linearly independent lattice points.

$$\lambda_1(\mathcal{L}) \leq \lambda_2(\mathcal{L}) \leq \dots \leq \lambda_n(\mathcal{L})$$

Lattice properties

recall, a lattice:

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- $B = (b_1, \dots, b_n) \in \mathbb{Z}^{n \times n}$: basis matrix, finite description of the lattice (not the lattice itself)
- the matrix-vector product Bz maps an integer coordinate vector $z \in \mathbb{Z}^n$ into a lattice point in $\mathbb{Z}^n$.

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- since columns of B are linearly independent (bcs they are a basis), every point of $\mathcal{L}(B)$ has a unique integer coordinate vector relative to the basis
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- since columns of B are linearly independent (bcs they are a basis), every point of $\mathcal{L}(B)$ has a unique integer coordinate vector relative to the basis
 - the lattice $\mathcal{L}(B)$ is the invariant set of points
 - the basis B is not unique
- $B \in \mathbb{Z}^{n \times n}$, n vectors, each with n coordinates $\implies$ requires $\mathcal{O}(n^2)$ integers.
 - only if we could have some better lattice structure to represent the lattice with fewer entries...

q-ary lattice

a full-rank lattice is q-ary if $q\mathbb{Z}^d \subseteq \mathcal{L} \subseteq \mathbb{Z}^d$

$\implies$ q-ary lattice is an infinite set of integer points, but membership depends only on the residue class modulo q

ie. if $x \in \mathcal{L}$, then $x + qz \in \mathcal{L} \ \forall z \in \mathbb{Z}^d$

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Let $A \in \mathbb{Z}^{n \times m}$ define the homomorphism

$$\begin{aligned}\psi_A : \mathbb{Z}^m &\longrightarrow \mathbb{Z}_q^n \\ z &\longmapsto Az \pmod{q}\end{aligned}$$

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q-ary lattice:

$$\mathcal{L}_q^\perp(A) = \ker(\psi_A) = \{z \in \mathbb{Z}^m \mid Az = 0 \pmod{q}\}$$

contains all vectors that are orthogonal modulo q to the rows of A .
The q-ary lattice generated by the rows of A :

$$\mathcal{L}_q(A) = \{y \in \mathbb{Z}^m \mid y = A^T z \pmod{q} \text{ for some } z \in \mathbb{Z}_q^n\}$$

Polynomial ring

At the NTRU paper, they introduced the usage of polynomial rings, which latter was ported over Ajtai work and other lattices, now is the general approach.

TODO NTRU lattices:

○

In $R = \mathbb{Z}_q[X]/(X^n + 1)$,

○ Addition:

$$a(x) + b(x) = \sum_{i=1}^d (a_i + b_i) X^i$$

○ Multiplication:

$$a(x) \cdot b(x) =$$

TODO

Timeline

- 1978 Knapsack cryptosystems attacked using lattice methods
- 1996 NTRU encryption
- 2005 Learning With Errors (LWE)
- 2008 Practical secure lattice-signature schemes
- 2009 Fully Homomorphic Encryption (FHE)
- 2016 NewHope (Ring-LWE KEM)
- 2017 Kyber (Module-LWE based) Dilithium (Module-LWE Module-SIS based)
- 2020s standardization: ML-KEM (Kyber) and ML-DSA (Dilithium)

Hard problems in lattices

- In elliptic curve cryptography, we rely on the hardness of the Discrete Logarithm problem:
given publicly known $g, p \in \mathbb{G}$, find privately known $x \in \mathbb{Z}$ such that $p = g^x$.
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find a shortest non-zero vector $x \in \mathcal{L}$, ie. that minimizes the Euclidean norm $\|x\|$, that is $\lambda_1(\mathcal{L}) = \|x\|$.

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Definition (CVP (Closest Vector Problem))

given a vector $y \in \mathbb{R}^n$ with $y \notin \mathcal{L}$, find a nonzero vector $x \in \mathcal{L}$ that is closest to y , ie. that minimizes the Euclidean norm $\|y - x\|$.

NTRU

Definition (NTRU key recovery problem)

Given $h(x)$, find ternary polynomials $f, g \in \mathcal{T}$ satisfying

$$f(x) \cdot h(x) = g(x) \pmod{q} \in R_q$$

Where $R_q = \mathbb{Z}_q[X]/(X^n - 1)$.

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- 1998, was first scheme using polynomial rings, which latter where approached as algebraically structured lattices
 - "lattice-based" bcs recovering the small secret polynomials can be formulated and attacked as a lattice problem
 - earlier lattice usage: in 1978, Knapsack cryptosystems targeted by lattice attacks
- following the approach, in 2002, Micciano modified Ajtai's function to work over polynomial rings getting efficiency improvements

Hard problems in lattices

Definition (SIS (Short Integer Solution))

given $q \in \mathbb{Z}$ and matrix $A \in \mathbb{Z}_q^{m \times n}$, find a non-zero short $z \in \mathbb{Z}^n$ s.th.

$$A \cdot z = 0 \pmod{q}$$

"short": $z \in \{B, B\}^n$ for some bound B , often $z \in \{-1, 0, 1\}^n$.

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The set of all solutions to the SIS problem forms the (q -ary) lattice

$$\Lambda_q^\perp(A) = \{z \in \mathbb{Z}^n \mid A \cdot z = 0 \pmod{q}\}$$

where the SIS problem becomes the SVP for $\Lambda_q^\perp(A)$

- LLL algorithm is used to solve SVP and CVP.
- Both SVP and CVP become computationally difficult (NP hard*) as the lattice dimension n grows.

Definition (Ring SIS (Ring Short Integer Solution))

given a uniformly random matrix $A \in {}^R R_q^n$, with elements $a_i \in R_q$,
find a non-zero integer vector $z \in R^n$ of norm $\|z\| \leq \beta$ such that

$$Az = \sum_i a_i \cdot z_i = 0 \in R_q$$

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Definition (MSIS (Module Short Integer Solution))

given a uniformly random matrix $A \in^R R_q^{m \times n}$, with columns
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Notice that Module-SIS is a Ring-SIS with $rank = 1$ ($m = 1$).

- without $\|z\| \leq \beta$, it is easy to find a solution via *Gaussian elimination*.
- $\beta < q$, because otherwise $z = (q, 0, \dots, 0) \in R^m$ would always be a valid solution.
- β and m must be large enough so that a solution is guaranteed to exist, usually $\beta \geq \sqrt{\hat{m}}$ and $m \geq \hat{m}$ where $\hat{m}$ is the smallest integer greater than $n \log q$.

Again: overall when doing lattice-based cryptography, we will be interested into keeping the values *short*, aka. of small norm.

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Again: overall when doing lattice-based cryptography, we will be interested into keeping the values *short*, aka. of small norm.
(example of decomposing values to keep small norms when doing operations in cryptographic protocols)

Hard problems in lattices

Definition (LWE (Learning With Errors))

choose a matrix $A \in {}^R \mathbb{Z}_q^{m \times n}$, a secret vector $s \in {}^R \mathbb{Z}_q^n$, and an error vector $e \in {}^R [-B, B]^m$. Let $b = As + e \pmod{q} \in \mathbb{Z}_q^m$. Given the pair (A, b) , the goal is to recover s .

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Search-LWE & Decisional-LWE problems.

For LWE samples, vector b is relatively close to exactly one vector in the LWE lattice:

$$\begin{aligned}\Lambda_q(A) &= \{A^T s \mid s \in \mathbb{Z}_q^n\} + q\mathbb{Z}^m \\ &= \{A^T s + qz \mid s \in \mathbb{Z}_q^n, z \in \mathbb{Z}^m\}\end{aligned}$$

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Observe: if $e = 0$, is a system of linear equations, which can be efficiently determined using Gaussian elimination over $\mathbb{Z}_q$. Hardness comes from the noise/error term $e \neq 0$ (and the norm bound).

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Observe: if $e = 0$, is a system of linear equations, which can be efficiently determined using Gaussian elimination over $\mathbb{Z}_q$. Hardness comes from the noise/error term $e \neq 0$ (and the norm bound). Similarly as with SIS, we also have Ring-LWE and Module-LWE variants of LWE.

Plain - Module - Ring

$$R_q = \mathbb{Z}_q[x]/(x^n + 1)$$

Assumption	Public linear relation	Underlying algebra
Plain LWE	$\mathbf{b} = A\mathbf{s} + \mathbf{e} \pmod{q}$	$A \in \mathbb{Z}_q^{m \times n}$
Module-LWE	$\mathbf{b} = A\mathbf{s} + \mathbf{e} \pmod{q}$	$A \in R_q^{k \times k}, \quad \mathbf{s}, \mathbf{e}, \mathbf{b} \in R_q^k$
Ring-LWE	$b = as + e \pmod{q}$	$a, s, e, b \in R_q$

Assumption	Public linear relation	Underlying algebra
Plain SIS	$A\mathbf{z} = \mathbf{0} \pmod{q}, \quad \mathbf{z} \neq \mathbf{0}, \quad \ \mathbf{z}\ \text{ small}$	$A \in \mathbb{Z}_q^{m \times n}, \quad \mathbf{z} \in \mathbb{Z}^n$
Module-SIS	$A\mathbf{z} = \mathbf{0} \pmod{q}, \quad \mathbf{z} \neq \mathbf{0}, \quad \ \mathbf{z}\ \text{ small}$	$A \in R_q^{m \times n}, \quad \mathbf{z} \in R^n$
Ring-SIS	$az = 0 \pmod{q}, \quad z \neq 0, \quad \ z\ \text{ small}$	$a \in R_q, \quad z \in R$

Dual lattices

SIS lattice:

$$\mathcal{L}^\perp(A) = \{z \in \mathbb{Z}^n \mid A \cdot z = 0 \pmod{q}\}$$

LWE lattice:

$$\begin{aligned}\mathcal{L}(A) &= \{A^T s \mid s \in \mathbb{Z}_q^n\} + q\mathbb{Z}^m \\ &= \{A^T s + qz \mid s \in \mathbb{Z}_q^n, z \in \mathbb{Z}^m\}\end{aligned}$$

Dual lattices

SIS lattice:

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Dual of a lattice:

$$\mathcal{L}^* = \{y \in \mathbb{R}^m \mid \langle y, x \rangle \in \mathbb{Z} \forall x \in \mathcal{L}\}$$

ie. all vectors that measure lattice points with integers.

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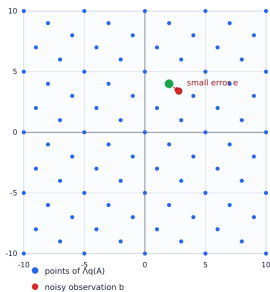
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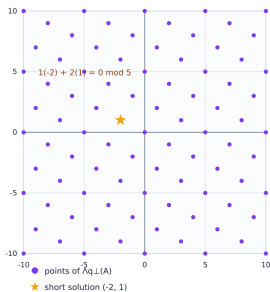
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SIS lattices and LWE lattices are dual to each other

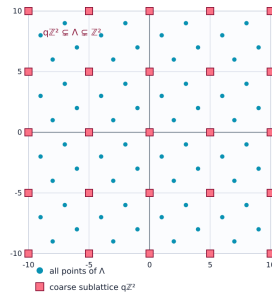
$$\mathcal{L}(A) = q\mathcal{L}^{\perp*}(A) \quad \text{ie.} \quad \frac{1}{q}\mathcal{L}(A) = \mathcal{L}^{\perp*}(A)$$

$q = 5$ Toy LWE: decode near $\Lambda_q(A)$ 

$$\Lambda_q(A) = \{s(1,2) + 5z : s \in \mathbb{Z}, z \in \mathbb{Z}^2\}$$

Toy SIS: $Az = 0 \pmod{q}$ 

$$\Lambda_{q\perp}(A) = \{z \in \mathbb{Z}^2 : ((1,2), z) = 0 \pmod{5}\}$$

q-ary lattice versus $q\mathbb{Z}^2$ 

$$\Lambda = 5\mathbb{Z}^2 + \mathbb{Z}(2,1) = \{5z + t(2,1) : z \in \mathbb{Z}^2, t \in \mathbb{Z}\}$$

Ajtai commitments

Definition (Ajtai commitment)

For a public random matrix $A \in \mathbb{Z}_q^{m \times n}$, embedding matrix $G \in \mathbb{Z}_q^{m \times l}$ and a short random vector $r \in \{-B, B\}^n$ (for bound B), an Ajtai commitment to the message $x \in \mathbb{Z}_q^l$ (bounded to B) is

$$c = Ar + Gx \pmod{q} \in \mathbb{Z}_q^n$$

Observe that this is closely related to the SIS problem ($Az = 0 \pmod{q}$), and it relies on its hardness.

ML-DSA signatures

Falcon signatures

LaBRADOR SNARK

Benchmarks

TODO prover and verifier speed, proof/signature size stark, labrador, sphinchs, falcon, ml-dsa, etc

Conclusions